

Mechanistic Interpretability & Activation Steering

Internal intervention mechanisms as artistic instruments of structured instability

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Status labels follow the project's canonical epistemic scheme.

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1 Executive summary

Mechanistic interpretability and activation steering are the only mechanisms in the AI SLOP research program that **literally write into a model's hidden state**. Everything else in the project — prompts, Temporary Minds, constraint governance, Suno lyric-jurisdiction tricks — operates on the input or on an external controller. Steering vectors, feature clamping, and activation patching operate on the residual stream itself, at a chosen layer, with a chosen magnitude, in real time. For a project whose core question is '*what problem can we give a generative system such that its attempt to solve it becomes the artwork*', this is the sharpest available instrument for manufacturing **representational conflict** with measurable dose.

The strongest verified findings: (a) contrastive steering vectors work as advertised on open models and their effects scale roughly monotonically with a coefficient before collapsing into incoherence — that cliff is itself an artistic failure surface^[3]; (b) sparse-autoencoder feature clamping produces coherent concept-takeover behavior (Golden Gate Claude being the canonical demonstration) and open SAE suites (Gemma Scope) plus the Neuronpedia platform make this reproducible by anyone with a consumer GPU^{[6][8]}; (c) the field's own benchmark, AxBench, is a discipline check — simple prompting beat SAE steering for concept injection, and a 2026 partial rebuttal recovered SAE performance only with careful per-feature feature selection and strength tuning^{[10][11]}. Steering is **controlled damage**, and damage is precisely the material this project studies.

Highest-value project connection: steering interventions are the most literal candidate for a **DAVID-operator analogue at the model-internal level** — an operator with named variables (layer, direction, coefficient schedule, token span) that preserves most of the model's competence while injecting a single persistent foreign pressure. Combined with the project's external state-controller rule (a controller that modulates the coefficient over generation time), it yields *PROCEDURAL* persistent instability with a real internal anchor — something no prompt-only method can claim honestly.

Honesty constraints honored throughout: no claim that prompting touches activations (it does not, unless hooks are attached); no claim of access to proprietary internals (all mechanisms here are demonstrated on open-weight models); feature labels are hypotheses until causally validated; nothing here is a safety-bypass technique, and refusal-direction manipulation is documented but deliberately excluded from the experiment set.

2 Technical background

A transformer language model carries a residual stream: a vector per token position that is read from and added to by every attention head and MLP block across layers. Mechanistic interpretability reverse-engineers what those vectors encode; activation steering writes into them at inference time. The shared formal core of nearly all steering methods is a single line: $\mathbf{x}' = \mathbf{x} + \alpha \cdot \mathbf{v}$, where $\mathbf{v}$ is a direction in activation space (a 'steering vector') and α is a coefficient. Methods differ in how $\mathbf{v}$ is obtained: mean difference of activations over contrast-pair prompts (CAA^[3], DiffMean), PCA of those differences (RepE/LAT^[4]), trained probes per attention head (ITI^[5]), decoder columns of a sparse autoencoder (SAE feature steering^[6]), or trained low-rank interventions (ReFT^[10]).

Sparse autoencoders deserve separate note. An SAE decomposes a residual-stream activation into a sparse weighted sum of learned feature directions; because the features are far more numerous than model dimensions and sparsely active, many of them are individually interpretable (a feature that fires

on the Golden Gate Bridge, on code errors, on sycophantic praise). Clamping one feature to a high value and reconstructing gives a surgical, if leaky, concept injection^{[6][7]}. Google DeepMind's Gemma Scope provides open SAEs for every layer of Gemma 2 2B/9B, and Gemma Scope 2 extends this to Gemma 3 with transcoders^{[8][9]}; Neuronpedia hosts these with interactive steering demos.

Verified tooling stack (all confirmed live as of 2026-09-19):

Tool	What it does	Status / license	Source
TransformerLens	Hook-based access/edit of any internal activation across 140+ architecture families; the standard workbench	Active (pushed 2026); MIT	github.com/TransformerLensOrg/TransformerLens
SAELens	Training and analysis of SAEs; v6 line adds JumpReLU, Matryoshka, Matching Pursuit variants	Very active (1.5k+ stars, pushed 2026-09-18); MIT	github.com/decoderresearch/SAELens
pyvene	Config-based interventions on any PyTorch model; static or trainable; serializable intervention objects shareable via HF	Active; Apache-2.0	github.com/stanfordnlp/pyvene
nnsight / NDIF	Deferred-execution intervention tracing; remote=True runs interventions on hosted models up to 405B+ without a local GPU	Active (v0.6, 2026); MIT	github.com/ndif-team/nnsight
Neuronpedia	Web platform: feature dashboards, search, interactive steering on hosted SAEs (Gemma Scope 1/2, Llama, Qwen releases)	Active; open source	neuronpedia.org
Gemma Scope	Open JumpReLU SAE suites for Gemma 2 (all layers/sublayers) and Gemma 3 (Scope 2)	Active; weights on HF (check per-repo terms)	huggingface.co/google/gemma-scope

Table 1 — Verified core tooling. Star counts and push dates checked via the GitHub API on 2026-09-19.

Starter-lead verification notes: **rattlesnakey/Awesome-Actionable-MI-Survey** is real, current (updated 2026-03), and unusually well-organized (Locate/Steer/Improve taxonomy tagging every paper by interpretable object and steering method)^[15]. **kmeng01/rome** is real but functionally frozen (last push 2024-04; the field moved to MEMIT and to EasyEdit-style suites). **koayon/awesome-sparse-autoencoders** is small (33 stars) and stale (2025-01); the Dakingrai list is stale (2024-11). The **yuzhaouoe/SAE-based-representation-engineering** list is actively maintained (2026-06) and is the best of the starter awesome-lists for this topic. The HF demo **huggingface/eiffel-tower-llama-demo** exists and is a complete, well-documented SAE-steering reference implementation (Llama 3.1 8B + ardit SAEs, forward-hook steering with configurable layer/feature/strength)^[16]. The dataset **future-probes/activation_steering** could not be verified via the HF API — treat as unverified until manually confirmed.

3 Mechanism catalog

Each entry follows the project template. Confidence tiers: **ESTABLISHED** / **DOCUMENTED** (peer-reviewed or widely replicated), **PLAUSIBLE INTERPRETATION** (consistent with evidence, not fully pinned), **PROJECT HYPOTHESIS** (our proposed artistic use), **SPECULATION**. Creative utility and mechanism confidence are scored separately (High / Medium / Low), per project rule: a beautiful accident may score high on the first and low on the second.

M1 — Contrastive Activation Addition (CAA)

SOURCE: Rimsky et al., 'Steering Llama 2 via Contrastive Activation Addition' (ACL 2024); code: github.com/nrimsky/CAA^[3]

SOURCE TYPE: Peer-reviewed paper + reference code

LICENSE / USAGE NOTES: Code MIT; needs local open-weight model and contrast-pair dataset (Anthropic's model-written-evals datasets are commonly reused)

WHAT IT ACTUALLY DOES: Averages last-token activation differences over paired prompts differing only in whether the target behavior is exhibited; adds $\alpha \cdot v$ to residual stream at a chosen layer during generation

CONTROLLABLE VARIABLES: Layer (single-layer application tolerated even at $|\alpha| > 1$), coefficient α (sign flips the behavior), token span (generation tokens only), vector construction set

WHAT STATE OR STRUCTURE IT PRESERVES: At moderate α : fluency, instruction-following, general competence — behavior shifts along one axis while grammar survives

WHAT IT CHANGES: Propensity toward the steered behavior/persona dimension (e.g., corrigibility, sycophancy)

FAILURE SURFACE: High α across many layers $\rightarrow$ incoherence; OOD fragility — vectors built on mismatched distributions generalize poorly^[14]

ORIGINAL PURPOSE: Safety-relevant behavior control without fine-tuning

POSSIBLE AI-SLOP MUTATION: Two opposed CAA vectors (e.g., 'euphoric' vs 'despairing', 'verbose scholar' vs 'broken telegraph') injected simultaneously at different layers — a literal internal contradiction the model must reconcile every token

EXPECTED OBSERVABLE ARTIFACT: Register oscillation within single sentences; metaphors that fight themselves; semantic whiplash with intact syntax — the 'governance error' made literal

AMPLIFICATION / ITERATION METHOD: Sweep $\alpha \blacksquare \times \alpha \blacksquare$ on a 2D grid; feed steered output back as next prompt while keeping vectors active (recursive contamination)

CONTROL OR ABLATION TEST: Random directions of matched norm (should produce generic degradation, not structured opposition); single-vector baselines

LIMITATION: Vectors capture population-average directions; fine, idiosyncratic concepts may not be linearly extractable

MECHANISM CONFIDENCE: High for the intervention mechanics; Medium for any specific concept pair behaving as imagined

APPLICABLE MEDIA: LLM; multimodal LLMs in principle (LMM MI literature is emerging^[15])

AI_SLOP CONNECTION: DAVID operator analogue (internal); mutation operator; experiment class

CONFIDENCE TIER: ESTABLISHED / DOCUMENTED

CREATIVE UTILITY: High

M2 — Activation Addition (ActAdd)

SOURCE: Turner et al., 'Activation Addition: Steering Language Models Without Optimization' (2023)^[2]

SOURCE TYPE: arXiv paper + public notebooks

LICENSE / USAGE NOTES: Open; trivially implementable with TransformerLens hooks

WHAT IT ACTUALLY DOES: Computes the difference between activations on two natural-language prompts (e.g., 'Love' – 'Hate') and adds it during generation — no dataset pairs, no averaging

CONTROLLABLE VARIABLES: Prompt pair, layer, α , injection window

WHAT STATE OR STRUCTURE IT PRESERVES: Same residual-stream geometry; single-site addition leaves downstream computation intact except through the added offset

WHAT IT CHANGES: Topic/attitude of generation toward the positive prompt's concept

FAILURE SURFACE: Concept bleed into unrelated content; coherence cliff at high α (documented in the original post and later surveys^[1])

ORIGINAL PURPOSE: Cheapest possible demonstration of linear representation control

POSSIBLE AI-SLOP MUTATION: Use semantically impossible prompt pairs ('the taste of Tuesday' – 'the sound of copper') so the extracted direction is itself an artifact of the model's confusion — steering by a nonsense compass

EXPECTED OBSERVABLE ARTIFACT: Low- α : subtle tonal contamination; mid- α : forced metaphorical interpretation of everything through the nonsense axis; high- α : collapse

AMPLIFICATION / ITERATION METHOD: Log-spaced α sweep; record the exact α where grammatical syntax outlives semantic coherence (the sweet-spot band is the artifact zone)

CONTROL OR ABLATION TEST: Same procedure with a grammatical but unrelated prompt pair; $\alpha = 0$ baseline; shuffled-vector control

LIMITATION: Single-prompt-pair vectors are noisy; concept purity is low compared to CAA

MECHANISM CONFIDENCE: High that the intervention does something directional; Low-Medium that a nonsense pair yields a meaningful axis — that is the experiment

APPLICABLE MEDIA: LLM

AI_SLOP CONNECTION: Mutation operator; new mechanism class ('steering by malformed probes')

CONFIDENCE TIER: ESTABLISHED (mechanism) / SPECULATION (malformed-pair variant)

CREATIVE UTILITY: High

M3 — SAE feature clamping / feature steering

SOURCE: Templeton et al., 'Scaling Monosemanticity' (Anthropic, 2024)^[6]; Gemma Scope (Google DeepMind)^[8]; Neuronpedia platform

SOURCE TYPE: Lab publication + open weights + live interactive demos

LICENSE / USAGE NOTES: Gemma Scope weights on HF (per-repo terms); SAELens (MIT) for loading; Neuronpedia free for exploration

WHAT IT ACTUALLY DOES: Encodes activations into SAE features, sets one feature's activation to a fixed value, decodes back, continues the forward pass — pins a single interpretable concept into every position

CONTROLLABLE VARIABLES: Feature index (searchable by concept on Neuronpedia), clamp value, layer = SAE site, duration (all tokens vs window)

WHAT STATE OR STRUCTURE IT PRESERVES: Fluency survives to surprisingly high clamp values (Golden Gate Claude stayed coherent while becoming the bridge)

WHAT IT CHANGES: The semantic center of gravity of the whole generation

FAILURE SURFACE: Reconstruction error from the SAE itself contaminates the intervention (matched-reconstruction controls are mandatory); at extreme values, monotone concept-zombification; AxBench showed naive feature steering underperforms prompting on concept-injection benchmarks^[10], partially rebutted with careful feature selection^[11]

ORIGINAL PURPOSE: Causal validation of interpretability claims; safety monitoring

POSSIBLE AI-SLOP MUTATION: Clamp two features with antagonistic meanings (e.g., 'silence' + 'alarm') to high values simultaneously; or clamp one feature while an external controller decays it slowly — watched entropy

EXPECTED OBSERVABLE ARTIFACT: Negotiation text: the model trying to satisfy both concepts produces genuine novelty (this is the AxBench 'cavalry' failure mode — 'The horse is ready to eat' — reframed as an artifact family^[11])

AMPLIFICATION / ITERATION METHOD: Grid over (feature A strength, feature B strength); map the phase diagram of coexistence vs dominance vs incoherence

CONTROL OR ABLATION TEST: Random unlabeled feature at same strength; dead-latent clamp (should do nothing); prompt-only concept injection comparison (AxBench discipline)

LIMITATION: Feature labels are hypotheses until causally tested; SAE reconstruction error is a hidden second intervention

MECHANISM CONFIDENCE: High that clamping shifts behavior; Medium that dual-clamp antagonism produces structured rather than mushy output

APPLICABLE MEDIA: LLM; vision-language models (SAEs with monosemantic features in VLMs demonstrated 2025^[15])

AI_SLOP CONNECTION: DAVID operator analogue; state-controller rule (time-varying clamp); validator (feature dashboards as readouts)

CONFIDENCE TIER: ESTABLISHED / DOCUMENTED

CREATIVE UTILITY: Very high

M4 — Activation patching / causal tracing

SOURCE: Meng et al. causal tracing (2022)^[12]; Zhang & Nanda patching conventions (2023); implemented in TransformerLens, pyvene, nnsight

SOURCE TYPE: Peer-reviewed papers + standard library primitives

LICENSE / USAGE NOTES: All tooling MIT/Apache; method is analysis-first

WHAT IT ACTUALLY DOES: Runs the model on a clean and a corrupted prompt, then grafts activation slices from one run into the other at chosen sites to localize what carries a behavior

CONTROLLABLE VARIABLES: Source/target run pair, patched site (layer × component × token), patch extent

WHAT STATE OR STRUCTURE IT PRESERVES: Everything outside the patched site — this is the most surgical intervention in the catalog

WHAT IT CHANGES: Usually diagnostic (which site restores/breaks the behavior); used generatively, it grafts a fragment of one run's computation into another

FAILURE SURFACE: Off-distribution grafts: a mid-layer residual vector from a different run is not a legal input to downstream layers, producing hybrid computations with no prompt-level analogue

ORIGINAL PURPOSE: Causal localization (which component is responsible for the fact/behavior)

POSSIBLE AI-SLOP MUTATION: Cross-domain grafting as composition: run A = a technical manual, run B = a lament; patch A's late-layer residuals into B mid-generation and let the model finish a text whose own machinery has been swapped out from under it

EXPECTED OBSERVABLE ARTIFACT: Register rupture at the patch boundary; 'scar tissue' sentences where the model re-coheres around the graft

AMPLIFICATION / ITERATION METHOD: Move the patch boundary token-by-token through the generation; each position is a different surgery — the sequence of outputs maps the graft's zone of influence

CONTROL OR ABLATION TEST: Self-patch (clean run into itself) must reproduce the original; random-direction patch as damage control

LIMITATION: Single-token effects dominate the literature; long-range graft consequences are under-characterized — open territory, meaning both opportunity and uncertainty

MECHANISM CONFIDENCE: High for diagnosis; Low-Medium for predictable generative outcomes of large grafts

APPLICABLE MEDIA: LLM

AI_SLOP CONNECTION: DAVID operator (surgical graft); experiment; connects to topic 9 (iterative encode-decode / artifact inheritance)

CONFIDENCE TIER: ESTABLISHED (diagnostic) / PROJECT HYPOTHESIS (generative grafting)

CREATIVE UTILITY: High

M5 — Persona vectors

SOURCE: Chen et al., 'Persona Vectors: Monitoring and Controlling Character Traits in Language Models' (Anthropic, 2025)^[13]; related: Lu et al., 'The Assistant Axis'

SOURCE TYPE: Lab research paper

LICENSE / USAGE NOTES: Method reproducible on open models; Anthropic artifacts not public, but the construction is standard contrastive difference-of-means

WHAT IT ACTUALLY DOES: Extracts directions corresponding to character traits (evil, sycophancy, hallucination-propensity, humor, optimism) from natural-language trait descriptions; can monitor, steer, or (via 'preventative steering' during fine-tuning) vaccinate against trait drift

CONTROLLABLE VARIABLES: Trait definition corpus, extraction layer, α , timing (inference vs training-time)

WHAT STATE OR STRUCTURE IT PRESERVES: General capabilities at moderate dose (benchmarked in the paper)

WHAT IT CHANGES: The 'who is speaking' layer — closer to Temporary Minds territory than any other mechanism in this catalog

FAILURE SURFACE: Trait entanglement: real traits are not independent axes, so pushing one drags neighbors; extreme persona clamping destabilizes the assistant frame entirely (cf. Assistant Axis drift findings)

ORIGINAL PURPOSE: Monitoring and preventing undesirable persona shifts, incl. emergent misalignment

POSSIBLE AI-SLOP MUTATION: A legitimately computed 'Temporary Mind' — extract a persona vector for one of the project's 26 minds from its procedural description, inject it, and compare against the prompt-only version. The delta between prompt-mediated and vector-mediated identity is measurable and itself an artwork about mediation

EXPECTED OBSERVABLE ARTIFACT: Vector-injected persona leaks into unprompted dimensions (the model's metaphors, its refusals, its punctuation habits) where prompt-only persona stays semantic

AMPLIFICATION / ITERATION METHOD: Dose sweep with a fixed probe battery measuring both target-trait expression and collateral trait drift

CONTROL OR ABLATION TEST: Prompt-only persona at matched surface instruction strength; orthogonal random vector

LIMITATION: Trait directions extracted from small corpora are fragile; cross-model transfer is poor

MECHANISM CONFIDENCE: Medium-High (peer-level lab work, but young literature)

APPLICABLE MEDIA: LLM

AI_SLOP CONNECTION: Temporary Mind (direct analogue); DAVID operator; experiment

CONFIDENCE TIER: ESTABLISHED / DOCUMENTED (method); PROJECT HYPOTHESIS (Temporary-Minds bridge)

CREATIVE UTILITY: Very high — the strongest single bridge to existing project machinery

M6 — ROME rank-one model editing

SOURCE: Meng et al., 'Locating and Editing Factual Associations in GPT' (NeurIPS 2022); repo [kmeng01/rome](#)^[12]

SOURCE TYPE: Peer-reviewed paper + reference implementation

LICENSE / USAGE NOTES: MIT; repo frozen since 2024-04 — use as reference, prefer maintained successors (MEMIT, EasyEdit suites) for new work

WHAT IT ACTUALLY DOES: Locates a factual association via causal tracing, then solves a closed-form rank-one update to an MLP weight matrix to rewrite one fact (e.g., 'Eiffel Tower is in Rome')

CONTROLLABLE VARIABLES: Target fact (subject–relation–object), edited layer(s), update strength

WHAT STATE OR STRUCTURE IT PRESERVES: Weights elsewhere untouched; designed to preserve all other behavior

WHAT IT CHANGES: The model's believed fact — permanently, until re-edited

FAILURE SURFACE: The famous one: edits don't generalize cleanly (rephrasings, implications of the edited fact often fail) and can distort nearby facts — the model holds a belief it cannot reason consistently about

ORIGINAL PURPOSE: Correcting factual errors without retraining

POSSIBLE AI-SLOP MUTATION: Install a deliberately false world-fact as a permanent internal axiom, then let prompt-level constraints reference the pre-edit world — the model's internals and its context now permanently disagree

EXPECTED OBSERVABLE ARTIFACT: Confabulation with conviction: the model defends the edited fact, invents supporting detail, and fails differently than a prompted liar does

AMPLIFICATION / ITERATION METHOD: Sequential edits building a small false cosmology; measure how contradictions between edits surface (edit interference)

CONTROL OR ABLATION TEST: Prompt-only false premise ('pretend the Eiffel Tower is in Rome') at matched strength; edit an unreferenced decoy fact

LIMITATION: Failure to generalize is well documented; this is a feature for the project (the inconsistency IS the artifact) but a bug for any truth claim

MECHANISM CONFIDENCE: High (heavily replicated, incl. ~20-line pyvene reproduction of the original tracing figure^[17])

APPLICABLE MEDIA: LLM

AI_SLOP CONNECTION: DAVID operator (persistent internal rewrite); state preservation study; experiment

CONFIDENCE TIER: ESTABLISHED / DOCUMENTED

CREATIVE UTILITY: Medium-High

M7 — Inference-Time Intervention (ITI)

SOURCE: Li et al., 'Inference-Time Intervention: Eliciting Truthful Answers from a Language Model' (NeurIPS 2023)^[5]

SOURCE TYPE: Peer-reviewed paper + code

LICENSE / USAGE NOTES: Open code; requires probe training per head

WHAT IT ACTUALLY DOES: Trains linear probes on attention-head outputs for a property (truthfulness), finds the K most informative heads, and shifts those heads' outputs along the probe direction at every generation step

CONTROLLABLE VARIABLES: Head count K, α , property dataset

WHAT STATE OR STRUCTURE IT PRESERVES: Most of the network; only top-K head outputs touched

WHAT IT CHANGES: A measurable behavioral axis (truthfulness benchmark gains were large in the paper)

FAILURE SURFACE: Multi-head intervention compounds; property labels are only as clean as the probe data

ORIGINAL PURPOSE: Truthfulness elicitation

POSSIBLE AI-SLOP MUTATION: Train probes for an aesthetic property with no ground truth ('sincerity', 'dread') and intervene anyway — the probe's label noise becomes the mechanism's signature

EXPECTED OBSERVABLE ARTIFACT: Consistent but unnameable tonal shift — the model drifts toward whatever the noisy probe actually encoded

AMPLIFICATION / ITERATION METHOD: Sweep K from 1 to 48 to watch intervention granularity trade against coherence

CONTROL OR ABLATION TEST: Random head selection at same K; probes trained on shuffled labels (must do nothing systematic)

LIMITATION: Per-head directions are entangled with many properties; 'truthfulness direction' framing has been critiqued as oversimplified

MECHANISM CONFIDENCE: Medium-High

APPLICABLE MEDIA: LLM

AI_SLOP CONNECTION: Mutation operator; experiment

CONFIDENCE TIER: ESTABLISHED / DOCUMENTED

CREATIVE UTILITY: Medium

M8 — Representation Engineering (RepE / LAT / LoRRA)

SOURCE: Zou et al., 'Representation Engineering: A Top-Down Approach to AI Transparency' (2023)^[4]

SOURCE TYPE: Paper + code (and a 2025 RepE survey systematizing the design space^[1])

LICENSE / USAGE NOTES: Open code; PCA-based pipeline

WHAT IT ACTUALLY DOES: Reads representations via Linear Artificial Tomography (contrast stimuli, PCA on differences), then controls via reading/contrast vectors or trains a low-rank adapter (LoRRA) to reproduce the steered activations in weights

CONTROLLABLE VARIABLES: Stimulus set, PCA component choice, α , per-layer targeting, adapter vs direct addition

WHAT STATE OR STRUCTURE IT PRESERVES: The top-down framing preserves interpretability of the control knob: you steer a named concept

WHAT IT CHANGES: High-level cognitive-function-like dimensions (honesty, emotion, morality framings in the paper)

FAILURE SURFACE: Named concepts are not guaranteed to be the actual encoded axes — the name is a hypothesis; survey documents strength/coherence trade-offs across the whole family^[1]

ORIGINAL PURPOSE: Transparency and control via top-down representations

POSSIBLE AI-SLOP MUTATION: Extract LAT directions for paired contradictory concepts from the SAME corpus and apply the second principal component instead of the first — steering by the residual axis nobody named

EXPECTED OBSERVABLE ARTIFACT: Coherent drift toward an unnamed attractor; post-hoc naming of the attractor becomes part of the artwork

AMPLIFICATION / ITERATION METHOD: Iterate: extract → steer → collect outputs → re-extract a new direction from the steered outputs (direction bootstrapping)

CONTROL OR ABLATION TEST: PC1 vs PC2 vs random subspace direction at matched norm

LIMITATION: PCA assumptions (linear, orthogonal) are known to be approximations; interpretive naming risk

MECHANISM CONFIDENCE: Medium-High for the machinery; the unnamed-axis variant is SPECULATION by design

APPLICABLE MEDIA: LLM; adaptable to multimodal representation spaces

AI_SLOP CONNECTION: DAVID operator; new mechanism class (unnamed-attractor steering)

CONFIDENCE TIER: ESTABLISHED (machinery) / SPECULATION (variant)

CREATIVE UTILITY: High

M9 — Difference-in-means & ReFT-r1 (AxBench family)

SOURCE: Wu et al., 'AxBench' (ICML 2025)^[10]; AxBench rank-1 leaderboard incl. HyperSteer, RePS^[18]

SOURCE TYPE: Benchmark paper + public dictionaries and leaderboard

LICENSE / USAGE NOTES: Dictionaries publicly released; usable via pyvene

WHAT IT ACTUALLY DOES: DiffMean: difference of mean activations between concept and non-concept examples, computed at scale over thousands of concepts. ReFT-r1: trained rank-1 intervention subspace per concept. Both released as SAE-scale dictionaries

CONTROLLABLE VARIABLES: Concept index (thousands), α , layer site, method (DiffMean vs trained variants)

WHAT STATE OR STRUCTURE IT PRESERVES: Benchmark-quantified fluency/instruction-following trade-off — this family is the best-instrumented for dose-response measurement

WHAT IT CHANGES: Concept incorporation strength, measurably, at scale

FAILURE SURFACE: The leaderboard itself documents the cliff: scores collapse differently per method as α grows; concept detection is far easier than concept steering — asymmetry worth exploiting

ORIGINAL PURPOSE: Standardized comparison of steering methods

POSSIBLE AI-SLOP MUTATION: Use the released dictionaries as a readymade concept arsenal: random-pair concept injection at benchmark-quantified doses = a compositional instrument with known failure calibration

EXPECTED OBSERVABLE ARTIFACT: Statistically predictable weirdness — you can select the α band where concept incorporation succeeds but instruction-following degrades (the benchmark's own harmonic-mean penalty zone)

AMPLIFICATION / ITERATION METHOD: Scripted traversal of the concept dictionary under fixed dose, generating a systematic corpus of controlled contaminations

CONTROL OR ABLATION TEST: The benchmark IS the control framework (fluency/concept/instruction ratings)

LIMITATION: Built on Gemma-2-2B/9B; LLM-judge scoring has known biases

MECHANISM CONFIDENCE: High

APPLICABLE MEDIA: LLM

AI_SLOP CONNECTION: Validator (benchmark as measurement instrument); experiment; mutation operator

CONFIDENCE TIER: ESTABLISHED / DOCUMENTED

CREATIVE UTILITY: Medium-High

M10 — Assistant Axis / activation capping

SOURCE: Lu et al., 'The Assistant Axis: Situating and Stabilizing the Default Persona of Language Models' (Anthropic Fellows, with Neuronpedia interactive demo)^[19]

SOURCE TYPE: Lab paper + live interactive demo

LICENSE / USAGE NOTES: Demo on Neuronpedia; method reproducible

WHAT IT ACTUALLY DOES: Identifies a direction along which the model's persona drifts away from 'assistant'; activation capping holds the model within the assistant region rather than pushing it anywhere new

CONTROLLABLE VARIABLES: Cap threshold, monitored direction(s), intervention site

WHAT STATE OR STRUCTURE IT PRESERVES: The default persona — this is a stabilizer, not a destabilizer

WHAT IT CHANGES: Suppresses drift into character-breaking trajectories (including harmful ones)

FAILURE SURFACE: Inverted for the project: capping is the control condition. Its interesting misuse is selective de-capping — removing the stabilizer locally while an external controller watches drift accumulate

ORIGINAL PURPOSE: Safety: keeping deployed models in-character

POSSIBLE AI-SLOP MUTATION: Drift cartography: deliberately let the persona axis run free under a conflicting-constraint load, log the trajectory in activation space, and use the logged path as a score for text generation

EXPECTED OBSERVABLE ARTIFACT: Persona erosion made visible as text structure — the moment of character-break becomes a formal event in the piece

AMPLIFICATION / ITERATION METHOD: Long-horizon generation with periodic axis projection logging; threshold alarms drive an external state machine

CONTROL OR ABLATION TEST: Capped vs uncapped under identical adversarial load

LIMITATION: Persona axes are population-level; individual drift is noisier

MECHANISM CONFIDENCE: Medium-High

APPLICABLE MEDIA: LLM

AI_SLOP CONNECTION: State-controller rule (drift-driven controller); validator; Temporary Mind (erosion study)

CONFIDENCE TIER: ESTABLISHED / DOCUMENTED (capping) / PROJECT HYPOTHESIS (drift cartography)

CREATIVE UTILITY: High

4 Failure surfaces as artistic material

The project treats failure surfaces as mechanisms. This topic offers four that are unusually well characterized — meaning the failure can be *aimed at* rather than stumbled into.

Failure surface	Documented behavior	Aiming parameters	Evidence tier
Coherence cliff	As steering coefficient α grows, generations pass from on-concept fluent $\rightarrow$ concept-forced $\rightarrow$ syntactically intact but semantically broken $\rightarrow$ token salad. The transitions are reproducible per model/layer	α (log-spaced sweep), layer choice, number of intervened layers (single-layer tolerates $ \alpha >1$ better than multi-layer)	ESTABLISHED ^[1]
Concept colonization	A clamped SAE feature progressively rewrites every topic as itself (Golden Gate: coherent but monomaniacal). Intermediate clamp values produce negotiation texts	Clamp value; feature choice; token window	ESTABLISHED ^[6]
Intervention asymmetry	Concept <i>detection</i> by representation methods is strong; concept <i>steering</i> is weak-to-moderate (AxBench). Reading minds is easier than writing them — the write channel is the lossy one	Method selection; benchmark scores as calibration	ESTABLISHED ^{[10][1]}
Edit inconsistency	ROME-style fact edits fail to propagate to rephrasings and implications — the model holds contradictory beliefs simultaneously and confabulates when pressed	Edit set design; probe battery targeting implication chains	ESTABLISHED ^[12]
Feature-label illusion	A Neuronpedia label is a hypothesis from top-activating examples; features fire on things outside their label, so naive steering by label mis-fires (AxBench 'hypertrophy house' failure)	Require causal validation before relying on a label; log mis-fires as artifacts	ESTABLISHED ^{[10][1]}

Table 2 — Characterized failure surfaces. Each row is an instrument: the 'failure' is reproducible, parameterizable, and documentable per the project's run-log template.

5 Promising experiments

E1 — The Contradiction Engine (dual-vector internal conflict)

HYPOTHESIS: Two antagonistic steering vectors injected simultaneously at different layers force a per-token reconciliation that produces structured register conflict unlike either single-vector baseline or a prompt-level contradiction

REPRESENTATIONAL TENSION: Vector A (e.g., 'clinical detachment') at an early-mid layer shapes semantic planning; vector B (e.g., 'grief') at a later layer shapes surface realization — the two jurisdictions cannot see each other directly

COMPETING CONSTRAINTS: Both vectors act on the same residual stream; the model must produce one coherent token distribution satisfying two incompatible pressures

WHY THE SYSTEM MAY STRUGGLE: The vectors were extracted independently; their sum is off the training manifold of coherent single-persona text

EXPECTED RESULT: Intra-sentence register rupture; metaphor systems that contradict their own framing; a measurable $\alpha \times \alpha$ phase diagram with a coexistence band, a dominance region, and an incoherence cliff

NEGATIVE CONTROL: Two random orthogonal directions of matched norms (expect generic mush, not structured opposition); prompt-only contradictory persona

ABLATION: A alone; B alone; A+B at the same layer (tests whether layer separation is load-bearing)

DOSE RESPONSE: 6×6 log-spaced grid over (α , α); fixed probe battery: fluency score, sentiment arc variance within sentences, human rating of 'structured vs mushy'

ITERATIVE VERSION: Steered output becomes next-round prompt while both vectors persist; measure whether the conflict compounds or equilibrates

FALSIFICATION CONDITION: Dual-vector outputs are statistically indistinguishable from single-vector mixtures or from random-direction controls on all probes

E2 — Concept colonization phase map (SAE clamp sweep)

HYPOTHESIS: Clamping a single Gemma-Scope feature across a strength sweep traces a reproducible trajectory from thematic mention → pervasive metaphor → identity takeover → incoherence, with sharp phase boundaries

REPRESENTATIONAL TENSION: The clamped feature demands expression at every token; the instruction-tuned conversational prior demands task relevance

COMPETING CONSTRAINTS: Feature clamp value vs. the model's coherence-maintenance capacity

WHY THE SYSTEM MAY STRUGGLE: Feature steering is 'controlled damage' — reconstruction error plus forced concept expression jointly push off-manifold^[11]

EXPECTED RESULT: Ordered phase sequence with identifiable boundary artifacts; the boundary zones (not the extremes) produce the richest text

NEGATIVE CONTROL: Dead-latent clamp (expect no effect); random unlabeled feature; matched-reconstruction control (clamp nothing but pass through the SAE) to isolate reconstruction damage from feature effect

ABLATION: Clamp at prompt tokens only vs generation tokens only vs all

DOSE RESPONSE: Clamp values across at least two orders of magnitude; 20 prompts × 8 strengths; score concept density, fluency, instruction adherence

ITERATIVE VERSION: Re-clamp at each generation round using the previous round's most-activated OTHER feature as the next clamp target (feature succession)

FALSIFICATION CONDITION: No ordered phase structure: effects are monotonic mush from the first nonzero clamp

E3 — The steered state machine (controller × internal intervention)

HYPOTHESIS: An external controller that modulates steering coefficient over time according to logged internal state (persona-axis projection, feature activations) produces path-dependent text that constant- α steering cannot

REPRESENTATIONAL TENSION: The controller's schedule (external memory, PROCEDURAL) vs. the model's autoregressive tendency to re-equilibrate after each intervention

COMPETING CONSTRAINTS: Controller pushes the model along a logged trajectory; model pulls toward its basin

WHY THE SYSTEM MAY STRUGGLE: Each intervention is a perturbation the next tokens must digest; repeated perturbation before digestion accumulates debt

EXPECTED RESULT: Text whose structure encodes the control trajectory: visible 'tides', hysteresis ($A \rightarrow B \rightarrow A$ does not return to A), scar passages after coefficient spikes

NEGATIVE CONTROL: Same total steering budget delivered at constant α ; random schedule with matched statistics

ABLATION: Controller with no state feedback (open loop) vs closed loop on logged projections

DOSE RESPONSE: Schedule amplitude and frequency sweep (slow drift vs spike-and-recover)

ITERATIVE VERSION: This IS the iterative version; long-horizon runs of 10k+ tokens with full activation logging

FALSIFICATION CONDITION: Closed-loop trajectories show no hysteresis and no structural dependence on schedule — path dependence is absent

E4 — Surgical graft (cross-run activation patching as composition)

HYPOTHESIS: Patching late-layer residual slices from run A (e.g., a technical manual passage) into run B (e.g., a lament) at a chosen token position produces a re-coherence artifact family with a measurable zone of influence around the graft point

REPRESENTATIONAL TENSION: Upstream computation believes it is writing A; downstream computation receives a vector encoding B's trajectory

COMPETING CONSTRAINTS: Continuity with generated prefix vs. consistency with grafted state

WHY THE SYSTEM MAY STRUGGLE: The grafted vector is off-distribution for the receiving layers in this context — a legal vector from an illegal history

EXPECTED RESULT: A 'scar' sentence at the graft boundary, then re-coherence whose style bears both ancestries; patch position sweep maps influence radius

NEGATIVE CONTROL: Self-patch (must reproduce original exactly); patch from an unrelated third run at matched norm

ABLATION: Patch size: single token position vs span; single layer vs adjacent layers

DOSE RESPONSE: Interpolated grafts: $(1-t) \cdot x_B + t \cdot x_A$ for t in $[0,1]$

ITERATIVE VERSION: Chain grafts: each output becomes run B for the next graft from a rotating cast of A-sources (artifact inheritance)

FALSIFICATION CONDITION: Grafts produce only local token corruption with no structured downstream signature distinguishable from random-vector damage

6 Cross-modal applications

Target medium	Transfer path	Status	Notes
Vision-language models	SAEs with monosemantic features in VLMs demonstrated (2025); steering vectors in LMMs tracked by the Awesome-LMMs-MI list (active, 2026-03)	DOCUMENTED, young	Concept clamping in a VLM could force visual descriptions through a foreign concept — bridge to topic 7
Image diffusion	Same $x' = x + \alpha \cdot v$ logic applies to diffusion latents and cross-attention maps — deferred to Topic 02 (inversion/latent manipulation)	NEXT PDF	CrossAttentionControl and PnPInversion leads already in the queue
Audio (Suno/API models)	No verified activation access to proprietary audio models; honest analogue is procedural: external controller + spectral operators	PROCEDURAL only	Internal steering for open audio diffusion models (e.g., teticio/latent-audio-diffusion line) is a Topic 06 item
GLSL / procedural	Not applicable directly; but the 'steering phase diagram' concept (maps of coexistence/dominance/incoherence regions) transfers as a visualization method for any parameterized system	METAPHOR	Useful as design language, not as mechanism
Controllers (TOPOS-SRE)	Strongest fit: any steering intervention exposes (layer, direction, α , window) as clean state variables for an external state machine	READY	E3 is written for this integration

Table 3 — Cross-modal transfer assessment.

7 What appears most useful for AI SLOP

1. Steering interventions are the project's first honest internal mechanism. Every prior mechanism in the registry is either prompt-level or controller-level. Activation steering is the first catalog entry where the mechanism physically resides inside the generative system during inference, with named, sweepable, ablatable variables. It upgrades DAVID-operator language from procedural metaphor to testable instrumentation — while the epistemic labels still apply (a steering vector's *artistic interpretation* remains a hypothesis).

2. The coherence cliff is a calibrated instrument, not an accident. Because the fluency-vs-concept trade-off is benchmarked (AxBench) and documented across methods, the project can *compose at the edge of collapse on purpose* and cite the dose. 'We steered Gemma-2-9B at α where concept incorporation survives but instruction-following measurably degrades' is a mechanism claim, not a vibe.

3. Persona vectors are the missing bridge to Temporary Minds. M5 proposes the cleanest experiment in this document: extract a vector-level Temporary Mind and measure the delta against its prompt-only twin. Whatever the outcome, the delta is publishable project knowledge — either vector mediation adds something prompting cannot (leakage into unprompted dimensions) or it does not (prompting sufficiency, consistent with AxBench's prompt baseline dominance).

4. Controller integration is already engineered. nnsight/NDIF removes the GPU barrier (remote interventions on hosted models up to 405B+), Neuronpedia removes the feature-discovery barrier, and

steering parameters map one-to-one onto the state variables an SRE controller already manages. E3 can be prototyped with a local Gemma-2-2B and SAELens in a weekend.

8 Weak / dubious ideas worth rejecting

Claim / idea	Verdict	Why
'Prompting manipulates the model's hidden activations'	REJECT as stated	Prompting changes inputs; activations change as a consequence of normal inference, which is not an intervention. Without hooks, no activation-level access exists. Keep the vocabulary clean
'SAE feature steering is a superior control method'	REJECT	AxBench: prompting outperforms SAE steering for concept injection; SAEs' value here is causal validation and failure-surface access, not control supremacy ^{[10][11]}
'Feature labels are ground truth'	REJECT	Labels come from top-activating examples; features misfire off-label. Label = hypothesis until causal test (the 'hypertrophy house' artifact is the counterexample ^[11])
'There is one direction for X' (generalized from refusal-direction results)	REJECT as generalization	Single-direction mediation is documented for refusal across 13 models ^[2] — that does not license single-direction claims for arbitrary concepts
'Steering reveals the model's true hidden self'	REJECT (mysticism)	Steering adds a vector. The output is a perturbed computation, not a confession. Project rule: no hidden-mind claims
'Refusal-direction erasure as artistic constraint removal'	OUT OF SCOPE	Documented mechanism (Arditi et al. 2024), deliberately excluded: the project is not safety-bypass research
kmeng01/rome as living infrastructure	DEPRECATE	Frozen since 2024-04; fine as reference implementation, wrong as a dependency for new work
future-probes/activation_steering dataset	UNVERIFIED	Not resolvable via HF API on 2026-09-19; do not cite until confirmed

Table 4 — Rejection register.

9 Open questions

- Do antagonistic dual-vector interventions produce phase structure (coexistence bands) or merely averaged mush? No published dose-response map for simultaneous opposed concepts was found — E1 targets exactly this gap.
- How does steered generation under persistent intervention evolve over long horizons (10k+ tokens)? The literature is dominated by short completions; drift, equilibration, and debt-accumulation dynamics are uncharacterized.
- Can a persona vector extracted from a procedural description (a Temporary Mind) capture anything the prompt version does not? The extraction corpus for a procedural identity is an unusual object — part prompt set, part score.
- Does grafting (activation patching) at generative scale produce re-coherence signatures distinct from simple corruption? Patching is characterized diagnostically, not compositionally.

- What is the right epistemic label for an artifact that appears only inside a narrow α band? The project's OBSERVED label covers behavior, but band-localized phenomena may deserve a named convention (e.g., 'windowed artifact').
- Feature succession (E2 iterative version): is the most-activated-other-feature rule a meaningful 'associative step' or an arbitrary jump? No prior work found on iterated feature-to-feature clamping.

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All URLs verified reachable or via official indexes on 2026-09-19 unless marked UNVERIFIED in Table 4.

11 Licensing / provenance appendix

Asset	License / terms	Verified	Provenance note
TransformerLens	MIT	2026-09-19 (GitHub API)	Created by Neel Nanda; maintained by Bryce Meyer & Jonah Larson
SAELens	MIT	2026-09-19 (GitHub API)	Decode Research; active development (v6 line)
pyvene	Apache-2.0	2026-09-19 (GitHub API)	Stanford NLP
nnsight	MIT (see repo)	2026-09-19	NDIF team, Northeastern University; remote execution requires free NDIF API key
kmeng01/rome	MIT	2026-09-19 (GitHub API)	Frozen since 2024-04 — reference only
nrimsky/CAA	MIT (see repo)	2026-09-19 (linked from paper)	Reference CAA implementation
Gemma Scope SAE weights	Gemma terms of use (check per-repo page on HF)	2026-09-19	Google DeepMind; Gemma model terms apply to derivatives
Neuronpedia	Open source; hosted platform	2026-09-19	Feature dashboards, steering demos; same org as SAELens
Awesome-Actionable-MI-Survey	Repo list (papers external)	2026-09-19	Active; best current map of actionable MI
SAE-based-representation-engineering (yuzhaouoe)	MIT	2026-09-19 (GitHub API)	Active curated list for SAE steering papers
eiffel-tower-llama-demo	See Space files	2026-09-19 (PROJECT.md)	Complete hook-based steering reference; uses andyrdt SAEs for Llama 3.1 8B
future-probes/activation_steering	—	UNVERIFIED	Not resolvable via HF API; excluded from mechanism claims

Table 5 — Licensing and provenance. GitHub metadata pulled from the public API on 2026-09-19; where a license was not re-confirmable programmatically, the entry says 'see repo'.

Research conducted 2026-09-19 via web sources, arXiv, GitHub API, and Hugging Face API. The ai_slop repository was read (AI_CONTEXT.md, README.md) for connection mapping only — no files were modified, added, or committed. Claims labeled ESTABLISHED trace to peer-reviewed venues or lab publications with public artifacts; PLAUSIBLE INTERPRETATION, PROJECT HYPOTHESIS, and SPECULATION are marked inline at point of use.